

13-Implementation-Roadmap.md

Healthcare Clinical Intelligence Platform

Master Implementation Roadmap

Version: 1.0

Status: Build Ready

Mission

Build a production-grade Healthcare Clinical Intelligence Platform using:

- Next.js
- FastAPI
- LangGraph
- Supabase
- Qdrant
- Neo4j
- Redis
- AWS

while maintaining:

- Enterprise Quality
 - Cost Efficiency
 - Governance
 - Observability
 - Scalability
-

Development Philosophy

Never build everything at once.

Build in layers.

Foundation

↓

Retrieval
↓
Agents
↓
Governance
↓
Observability
↓
Scale

Repository Structure

Frontend

hcip-frontend

Tech:

- Next.js
 - TypeScript
 - ShadCN
-

Backend

hcip-backend

Tech:

- FastAPI
 - LangGraph
-

Infrastructure

hcip-infra

Tech:

- Docker
 - Terraform
 - Kubernetes
-

Evaluation

hcip-evaluation

Tech:

- RAGAS
 - DeepEval
-

Phase 0

Foundation Setup

Duration:

1 Week

Deliverables

Project Structure

Docker Setup

Supabase

Qdrant

Neo4j

Redis

Local Environment

CI/CD Skeleton

Success Criteria

All services running locally.

Phase 1

Ingestion Platform

Duration:

2 Weeks

Build

Upload Service

Document Service

Docling Integration

OCR Pipeline

Metadata Extraction

Chunking Engine

Embedding Service

Versioning

Approval Workflow

Deliverables

Upload PDF

↓

Chunk

↓

Embed

↓
Store

Success Criteria

100+ documents successfully indexed.

Phase 2

Retrieval Platform

Duration:

2 Weeks

Build

Hybrid Search

BM25

Qdrant Retrieval

Metadata Filters

Parent Child Retrieval

Query Rewriting

Multi Query Retrieval

Reranking

Deliverables

Enterprise Retrieval Layer

Success Criteria

Context Precision > 85%

Phase 3

GraphRAG

Duration:

1 Week

Build

Entity Extraction

Knowledge Graph

Neo4j Integration

Graph Retrieval

Deliverables

GraphRAG Support

Success Criteria

Drug-Disease Graph Queries Working

Phase 4

Self-Correcting RAG

Duration:

2 Weeks

Build

Confidence Engine

Evidence Validator

Corrective Retrieval

Retry Logic

Source Diversity

Deliverables

Self-Healing Retrieval

Success Criteria

Hallucination Reduction

Phase 5

LangGraph Agent System

Duration:

2 Weeks

Build

Planner Agent

Retriever Agent

Validator Agent

Safety Agent

Response Agent

Memory Agent

Citation Agent

Deliverables

Multi-Agent Platform

Success Criteria

All Agents Working Together

Phase 6

Memory Architecture

Duration:

1 Week

Build

User Memory

Department Memory

Organization Memory

Agent Memory

Memory Governance

Deliverables

Enterprise Memory Layer

Success Criteria

Personalized Responses

Phase 7

Frontend Platform

Duration:

3 Weeks

Build

Authentication

Dashboard

AI Assistant

Knowledge Bases

Document Management

Memory Module

Admin Portal

Deliverables

Production UI

Success Criteria

Complete User Journey

Phase 8

Evaluation Platform

Duration:

1 Week

Build

RAGAS

DeepEval

Golden Dataset

Regression Testing

Deliverables

Evaluation Dashboard

Success Criteria

Automated Quality Reports

Phase 9

Observability Platform

Duration:

1 Week

Build

Langfuse

OpenTelemetry

Prometheus

Grafana

Loki

Deliverables

Full Tracing

Success Criteria

100% Request Visibility

Phase 10

Governance Platform

Duration:

1 Week

Build

RBAC

Approval Workflows

Audit Logs

Compliance Rules

Deliverables

Enterprise Governance

Success Criteria

Zero Unauthorized Access

Phase 11

Production Deployment

Duration:

1 Week

Build

Docker

Cloudflare

AWS

Backups

Monitoring

Autoscaling

Deliverables

Production Environment

Success Criteria

Stable Deployment

MVP Scope

Includes

Document Upload

Hybrid Search

Agentic RAG

Citations

Confidence Scores

Admin Portal

Observability

V1 Scope

Adds

GraphRAG

Memory

Self-Correcting RAG

Governance

Evaluation

V2 Scope

Adds

FHIR

EHR Integration

Clinical Workflows

Advanced Automation

Claude Code Workflow

Use Claude Code For

Backend

LangGraph

FastAPI

Retrieval

Infrastructure

Testing

Evaluation

Antigravity Workflow

Use Antigravity For

Next.js

UI Components

Dashboards

Admin Screens

Tables

Forms

Daily Development Flow

Architecture

↓

Claude Code

↓

Review
↓
Testing
↓
Commit
↓
Deploy

Weekly Milestones

Week 1

Infrastructure

Week 2-3

Ingestion

Week 4-5

Retrieval

Week 6

GraphRAG

Week 7-8

Self Correcting RAG

Week 9-10

Agents

Week 11

Memory

Week 12-14

Frontend

Week 15

Evaluation

Week 16

Observability

Week 17

Governance

Week 18

Production Deployment

Build Order Priority

Priority 1

Retrieval Quality

Priority 2

Agent Quality

Priority 3

Governance

Priority 4

Observability

Priority 5

Frontend Polish

Portfolio Value

By completion you will demonstrate:

- Advanced RAG
 - GraphRAG
 - Self-Correcting RAG
 - Multi-Agent Systems
 - LangGraph
 - Healthcare AI
 - Enterprise SaaS
 - AI Governance
 - AI Observability
 - AI Evaluation
 - Cloud Infrastructure
-

Final Success Criteria

The platform should be capable of:

10,000+ Queries / Day

100,000+ Documents

10M+ Chunks

99.9% Availability

<5s Responses

<5% Hallucination Rate

>95% Citation Coverage

while maintaining enterprise-grade security, governance, observability, and cost efficiency.